# ReadyShell Overlay Inventory Report (v0.2D)

Artifact-backed inventory for the current ReadyShell overlay system. This report is generated from the live build outputs, the ReadyShell linker map, and the active D71 disk image instead of hand-maintained notes.

## Executive Summary

| PROFILE | DISK SOURCE | OVERLAY WINDOW | WINDOW SIZE | RESIDENT HEAP |
|---|---|---|---|---|
| `precog-dual-d71` | `releases/0.2/precog-dual-d71/readyos-v0.2d-dual-d71_1.d71` | `$BE00-$C5FF` | **14336 bytes** | **1274 bytes** |

Overlays 1-8 are boot-loaded once and cached in fixed full-window REU slots. Bank `0x40` holds overlays `1`, `2`, `3`, and `5`; bank `0x41` holds overlays `4`, `6`, `7`, and `8`. Bank `0x48` is shared for the external-command registry, overlay metadata, pause state, command handoff scratch, and the REU-backed ReadyShell value arena.

### Runtime Memory Map

- **Resident app window:** `$1000-$C5FF` (46592 bytes)
- **Overlay load bytes:** `$BDFE-$BDFF`
- **Overlay execution window:** `$BE00-$C5FF` (14336 bytes)
- **Resident BSS:** `$87BC-$8982` (503 bytes)
- **Resident heap:** `$8904-$BDFD` (1274 bytes)
- **High RAM runtime:** `$CA00-$CFFF` (1536 bytes)

### REU Layout

- **Shared cache bank 1:** `$400000-$40FFFF` (65536 bytes)
- **Overlay 1 parse slot:** `$400000-$4037FF` (14336 bytes)
- **Overlay 2 exec slot:** `$403800-$406FFF` (14336 bytes)
- **Overlay 3 command slot:** `$407000-$40A7FF` (14336 bytes)
- **Overlay 5 command slot:** `$40A800-$40DFFF` (14336 bytes)
- **Cache bank 1 free tail:** `$40E000-$40FFFF` (8192 bytes)
- **Shared cache bank 2:** `$410000-$41FFFF` (65536 bytes)
- **Overlay 4 command slot:** `$410000-$4137FF` (14336 bytes)
- **Overlay 6 command slot:** `$413800-$416FFF` (14336 bytes)
- **Overlay 7 command slot:** `$417000-$41A7FF` (14336 bytes)
- **Overlay 8 command slot:** `$41A800-$41DFFF` (14336 bytes)
- **Cache bank 2 free tail:** `$41E000-$41FFFF` (8192 bytes)
- **Debug trace ring:** `$43F000-$43F20F` (528 bytes)
- **Command scratch:** `$480000-$487FFF` (32768 bytes)
- **Command registry header:** `$488010-$488017` (8 bytes)
- **Command descriptor table:** `$488020-$48807F` (96 bytes)
- **Overlay state table:** `$48808D-$48808E` (108 bytes)
- **Shared metadata:** `$4880F0-$4880FB` (12 bytes)
- **Pause flag:** `$4880FC` (1 byte)
- **Heap metadata:** `$488000-$4880FF`
- **Heap arena:** `$488100-$48FEFF` (32256 bytes)

## Shared REU Bank Visual

### Bank `0x40` : Overlay Cache

**Overlay 1 parse slot** `$400000-$4037FF`
Full `0x3000` Byte window image for `rsparser.prg`. Live payload is `13005` bytes, but the cache stores the whole overlay window so writable overlay data survives phase switching.

**Overlay 2 exec slot** `$403800-$406FFF`
Full `0x3000` Byte window image for `rsvm.prg`. Live payload is `14033` bytes.

**Overlay 3 command slot** `$407000-$40A7FF`
Full window snapshot for `rsdrvilst.prg`, serving both `DRVI` and `LST`.

**Overlay 5 command slot** `$40A800-$40DFFF`
Full window snapshot for `rsstv.prg`.

**Free tail** `$40E000-$40FFFF`
`8192` bytes left free after the four fixed slots in bank `0x40`.

### Bank `0x41` : Overlay Cache

**Overlay 4 command slot** `$410000-$4137FF`
Full window snapshot for `rsldv.prg`.

**Overlay 6 command slot** `$413800-$416FFF`
Full window snapshot for `rsfops.prg`.

**Overlay 7 command slot** `$417000-$41A7FF`
Full window snapshot for `rscat.prg`.

**Overlay 8 command slot** `$41A800-$41DFFF`
Full window snapshot for `rscopy.prg`.

**Free tail** `$41E000-$41FFFF`
`8192` bytes left free after the four fixed slots in bank `0x41`.

### Bank `0x48` : Shared Shell State

**Command registry** `$488010-$4880EB`
Registry header, external-command descriptors, and per-overlay load/cache state stored in REU metadata so new external commands do not consume resident BSS.

**Shared metadata + pause** `$4880F0-$4880FC`
Overlay cache metadata record plus the shared pause bit used by the resident line printer and `MORE`.

**Command scratch + value arena** `$480000-$48FEFF`
Command handoff scratch, REU heap metadata, and the persistent value arena for REU-backed strings, arrays, and objects.

## Command Scratch And Value Arena Usage

The command-scratch window is a single shared REU work area, so only the active command overlay owns it at any moment even though all command overlays are now preloaded into REU. The value arena is persistent session state in bank `0x48`. `LDV` populates it directly, while `STV` can serialize values that already live there.

| Commands | Overlay | Command Scratch | Value Arena | How The REU Region Is Used |
|---|---|---|---|---|
| PRT, MORE, TOP, SEL, GEN, TAP | `2` / `rsvm` | No direct use | Indirect only | Run inside the shared execution core. They do not stage command local data in `$480000-$487FFF`; any REU-backed values are handled through the normal overlay 2 value/runtime paths. |
| DRVI | `3` / `rsdrvilst` | No | No | Reads drive header/status data and builds its output object in transient overlay local RAM. |
| LST | `3` / `rsdrvilst` | Yes | No | Spools 28 byte directory records through `$480000-$487FFF` so `BEGIN/ITEM` can walk the listing without keeping the directory channel open. |
| LDV | `4` / `rsldv` | Yes | Yes, writes persistent values | Reads the RSV1 file into `$480000-$487FFF`, validates its header, then materializes strings, arrays, and objects into the REU heap arena `$488100-$48FEFF`. |
| STV | `5` / `rsstv` | Yes | Yes, reads existing pointer values | Uses `$480000-$4000FF` for session metadata and `$480100-$487FFF` for the outgoing RSV1 payload. When serializing pointer-backed values, it dereferences them from the persistent REU heap arena before flattening them into scratch. |
| DEL, REN | `6` / `rsfops` | No | No | Issue DOS scratch/rename commands directly through command-channel I/O with no REU staging. |
| PUT, ADD | `6` / `rsfops` | Yes | No | Use `$480000-$48001F` for session metadata and `$480020-$487FFF` as a text spool for new/appended SEQ content before writing it back to disk. |
| CAT | `7` / `rscat` | Yes | No | Uses `$480000-$480FFF` as a line-record table and `$480800-$487FFF` as the line-data spool so `ITEM` can replay file lines after the initial read pass. |
| COPY | `8` / `rscopy` | No | No | Uses its overlay-local 128 byte transfer buffer plus direct DOS copy/streamed file I/O. It does not use the shared command scratch or the persistent value arena. |

## Static Audit Checks

These checks are derived from the current source layout, not hand-maintained prose. The report generator now fails if the ReadyShell REU metadata packing or overlay registry capacities drift out of sync.

- Registry capacity check: `rs_cmd_registry.c` sends `10` external command descriptors into `16` reserved descriptor slots and `6` overlay-state rows into `6` reserved state slots.
- Metadata-page packing check: the full ReadyShell metadata block fits inside `$488000-$4880FF`. Header uses `$488010-$488017`, descriptor rows reserve `$488020-$48807F` with live rows ending at `$48805B`, state rows reserve `$488080-$4880EB` with live rows ending at `$4880EB`, shared metadata uses `$4880F0-$4880FB`, and the pause flag sits at `$4880FC`.
- Non-overlap check: command scratch ends at `$487FFF` and REU heap metadata begins at `$488000`; the state table ends at `$4880EB` and shared metadata begins at `$4880F0`; shared metadata ends at `$4880FB` and the pause flag is `$4880FC`; the value arena begins at `$488100`.
- Cache-slot audit: ReadyShell caches overlays `1`-`8`. Bank `0x40` carries overlays `1`, `2`, `3`, and `5`; bank `0x41` carries overlays `4`, `6`, `7`, and `8`. Every slot is a full `14336`-byte overlay-window snapshot.
- Command-source audit: `DRVI` builds output only in overlay-local RAM; `LST` writes 28 byte directory records into shared scratch; `LDV` streams RSV1 payloads into scratch and materializes persistent values into the REU heap arena; `STV` serializes into scratch and dereferences pointer-backed values from the arena; `DEL` and `REN` issue direct DOS commands without REU staging; `PUT` and `ADD` use scratch metadata plus a text spool; `CAT` uses a scratch record table plus line-data spool; `COPY` stays overlay-local with `g_copy_buf[128]`.

## Overlay Inventory

| Ovl | Role | Build PRG | Disk Name | PRG Bytes | Disk Blocks | Live Bytes | Window Use | Commands | REU |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Parser / Lexer | `rsparser.prg` | `rsparser` | 13007 | 52 | 13005 | 90.7% | None directly; parse phase support. | bank `0x40` slot `$400000-$4037FF` |
| 2 | Execution Core | `rsvm.prg` | `rsvm` | 14035 | 56 | 14033 | 97.9% | PRT, MORE, TOP, SEL, GEN, TAP and the shared execution paths that command overlays return to. | bank `0x40` slot `$403800-$406FFF` |
| 3 | Drive Info + Directory Listing | `rsdrvilst.prg` | `rsdrvilst` | 11274 | 45 | 11272 | 78.6% | DRVI, LST | bank `0x40` slot `$407000-$40A7FF` |
| 4 | Load Value | `rsldv.prg` | `rsldv` | 11886 | 47 | 11884 | 82.9% | LDV | bank `0x41` slot `$410000-$4137FF` |
| 5 | Store Value | `rsstv.prg` | `rsstv` | 10207 | 41 | 10205 | 71.2% | STV | bank `0x40` slot `$40A800-$40DFFF` |
| 6 | File Delete / Rename / Write | `rsfops.prg` | `rsfops` | 14335 | 57 | 14333 | 100.0% | DEL, REN, PUT, ADD | bank `0x41` slot `$413800-$416FFF` |
| 7 | File Read | `rscat.prg` | `rscat` | 8008 | 32 | 8078 | 56.3% | CAT | bank `0x41` slot `$417000-$41A7FF` |
| 8 | File Copy | `rscopy.prg` | `rscopy` | 6601 | 26 | 6599 | 46.0% | COPY | bank `0x41` slot `$41A800-$41DFFF` |

## Command Topology

This view is command-centric rather than file-centric. Shared overlays stand out here because multiple command pills sit inside one overlay card.

**Overlay 2** — SHARED EXECUTION CORE
`rsvm`
Shared execution paths that command overlays return to.
`PRT` `MORE` `TOP` `SEL` `GEN` `TAP`

**Overlay 3** — SHARED COMMAND OVERLAY
`rsdrvilst`
Shared command overlay for DRVI and LST.
`DRVI` `LST`

**Overlay 4** — SINGLE COMMAND OVERLAY
`rsldv`
Single-command overlay for LDV.
`LDV`

**Overlay 5** — SINGLE COMMAND OVERLAY
`rsstv`
Single-command overlay for STV.
`STV`

**Overlay 6** — SHARED COMMAND OVERLAY
`rsfops`
Shared command overlay for DEL, REN, PUT, and ADD.
`DEL` `REN` `PUT` `ADD`

**Overlay 7** — SINGLE COMMAND OVERLAY
`rscat`
Single-command overlay for CAT.
`CAT`

**Overlay 8** — SINGLE COMMAND OVERLAY
`rscopy`
Single-command overlay for COPY.
`COPY`

## Resident Program

- **Build PRG:** `readyshell.prg`
- **Disk name:** `readyshell`
- **Disk footprint:** 30478 bytes, 120 D71 blocks
- **Resident sources:** `readyshell.c`, `rs_token.c`, `rs_bc.c`, `rs_errors.c`, `rs_cmd_registry.c`, `rs_vm_c64.c`, `rs_overlay_c64.c`, `rs_platform_c64.c`, `tui_nav.c`, `reu_mgr_dma.c`, `resume_state_ctx.c`, `resume_state_core.c`
- **Resident asm/runtime:** `rs_runtime_c64.s`
- **Role:** Resident app shell loop plus vm/overlay runtime. Command tokens resolved here, then dispatched to overlay 2 or command overlays.
- **Current resident segments:** `CODE $7224`, `RODATA $041A`, `DATA $0047`, `INIT $001C`, `ONCE $0038`, `BSS $01F7`

## Per-Overlay Detail

### Overlay 1: Parser / Lexer

Lexer, parser, AST construction, and parse cleanup.

- **Build PRG:** `rsparser.prg`
- **Disk staging PRG:** `obj/rsparser.prg`
- **Disk filename:** `rsparser`
- **Source files:** `rs_lexer.c`, `rs_parse.c`, `rs_parse_support.c`, `rs_parse_free.c`
- **Commands:** None directly; parse phase support.
- **Live window range:** `$BE00-$C0CC` (13005 bytes)
- **Window share:** 90.7% used, 1331 bytes free
- **Disk footprint:** 13007 bytes, 52 D71 blocks
- **REU policy:** Boot-loaded from disk during shell startup, then restored from a fixed REU cache slot.
- **RAM notes:** Lives entirely inside the shared overlay window while active.

### Overlay 2: Execution Core

Values, variables, formatting, pipes, and shared execution helpers.

- **Build PRG:** `rsvm.prg`
- **Disk staging PRG:** `obj/rsvm.prg`
- **Disk filename:** `rsvm`
- **Source files:** `rs_vars.c`, `rs_value.c`, `rs_format.c`, `rs_cmd.c`, `rs_pipe.c`
- **Commands:** PRT, MORE, TOP, SEL, GEN, TAP and the shared execution paths that command overlays return to.
- **Live window range:** `$BE00-$C4D0` (14033 bytes)
- **Window share:** 97.9% used, 303 bytes free
- **Disk footprint:** 14035 bytes, 56 D71 blocks
- **REU policy:** Boot-loaded from disk during shell startup, then restored from a fixed REU cache slot.
- **RAM notes:** Includes `rs_vm_fmt_buf[128]` and `rs_vm_line_buf[384]` inside the overlay image.

### Overlay 3: Drive Info + Directory Listing

Shared command overlay for DRVI and LST.

- **Build PRG:** `rsdrvilst.prg`
- **Disk staging PRG:** `obj/rsdrvilst.prg`
- **Disk filename:** `rsdrvilst`
- **Source files:** `rs_cmd_lst_c64.c`, `rs_cmd_drvi_c64.c`
- **Commands:** DRVI, LST
- **Live window range:** `$BE00-$BA07` (11272 bytes)
- **Window share:** 78.6% used, 3064 bytes free
- **Disk footprint:** 11274 bytes, 45 D71 blocks
- **REU policy:** Boot-loaded from disk during shell startup, then restored from a fixed REU cache slot.
- **RAM notes:** Shares the inter-command REU handoff area at 0x480000-0x487FFF.

### Overlay 4: Load Value

Single-command overlay for LDV.

- **Build PRG:** `rsldv.prg`
- **Disk staging PRG:** `obj/rsldv.prg`
- **Disk filename:** `rsldv`
- **Source files:** `rs_cmd_ldv_c64.c`
- **Commands:** LDV
- **Live window range:** `$BE00-$BC6B` (11884 bytes)
- **Window share:** 82.9% used, 2452 bytes free
- **Disk footprint:** 11886 bytes, 47 D71 blocks
- **REU policy:** Boot-loaded from disk during shell startup, then restored from a fixed REU cache slot.
- **RAM notes:** Uses the shared handoff region plus the REU-backed value arena in bank 0x48 when hydrating pointer-backed values.

### Overlay 5: Store Value

Single-command overlay for STV.

- **Build PRG:** `rsstv.prg`
- **Disk staging PRG:** `obj/rsstv.prg`
- **Disk filename:** `rsstv`
- **Source files:** `rs_cmd_stv_c64.c`
- **Commands:** STV
- **Live window range:** `$BE00-$BD0C` (10205 bytes)
- **Window share:** 71.2% used, 4131 bytes free
- **Disk footprint:** 10207 bytes, 41 D71 blocks
- **REU policy:** Boot-loaded from disk during shell startup, then restored from a fixed REU cache slot.
- **RAM notes:** Uses the shared handoff region plus the REU-backed value arena in bank 0x48 when serializing pointer-backed values.

### Overlay 6: File Delete / Rename / Write

Shared command overlay for DEL, REN, PUT, and ADD.

- **Build PRG:** `rsfops.prg`
- **Disk staging PRG:** `obj/rsfops.prg`
- **Disk filename:** `rsfops`
- **Source files:** `rs_cmd_delren_c64.c`, `rs_cmd_putadd_c64.c`
- **Commands:** DEL, REN, PUT, ADD
- **Live window range:** `$BE00-$C5FC` (14333 bytes)
- **Window share:** 100.0% used, 3 bytes free
- **Disk footprint:** 14335 bytes, 57 D71 blocks
- **REU policy:** Boot-loaded from disk during shell startup, then restored from a fixed REU cache slot.
- **RAM notes:** Keeps file-operation staging and transient command state in overlay-local code plus the shared REU scratch region.

### Overlay 7: File Read

Single-command overlay for CAT.

- **Build PRG:** `rscat.prg`
- **Disk staging PRG:** `obj/rscat.prg`
- **Disk filename:** `rscat`
- **Source files:** `rs_cmd_cat_c64.c`
- **Commands:** CAT
- **Live window range:** `$BE00-$AD80` (8078 bytes)
- **Window share:** 56.3% used, 6258 bytes free
- **Disk footprint:** 8080 bytes, 32 D71 blocks
- **REU policy:** Boot-loaded from disk during shell startup, then restored from a fixed REU cache slot.
- **RAM notes:** Uses overlay-local file I/O logic plus shared REU scratch when line staging is needed.

### Overlay 8: File Copy

Single-command overlay for COPY.

- **Build PRG:** `rscopy.prg`
- **Disk staging PRG:** `obj/rscopy.prg`
- **Disk filename:** `rscopy`
- **Source files:** `rs_cmd_copy_c64.c`
- **Commands:** COPY
- **Live window range:** `$BE00-$A7C6` (6599 bytes)
- **Window share:** 46.0% used, 7737 bytes free
- **Disk footprint:** 6601 bytes, 26 D71 blocks
- **REU policy:** Boot-loaded from disk during shell startup, then restored from a fixed REU cache slot.
- **RAM notes:** Uses an overlay-local 128 byte transfer buffer plus direct DOS copy or streamed file I/O. It does not use the shared REU scratch or value arena.

## Observations

- Overlay 2 is effectively full at 14033 bytes of 14336 (97.9%).
- Overlay 1 is also large at 13005 bytes (90.7%).
- The resident heap below the overlay load address is only 1274 bytes, so large working sets depend on overlays and REU-backed storage.
- ReadyShell now uses two fixed REU cache banks: `0x40` for overlays `1`, `2`, `3`, and `5`, and `0x41` for overlays `4`, `6`, `7`, and `8`.
- Bank `0x40` leaves 8192 bytes free; bank `0x41` leaves 8192 bytes free.
- External commands now pay a one-time boot preload cost instead of a repeated disk-load cost during each command call.
- Overlay 2 owns the shared formatting buffers, which is why it consumes almost the entire overlay window.